

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2019- 2020 (Odd Semester)
Innovative Practices Description

UNIT I INFORMATION THEORY

Degree, Semester& Branch: V Semester B.E. ECE B

Course Code & Title: EC8501 Digital Communication

Name of the Faculty member: Mr.R.Deiva Nayagam

Name of the Topic: Huffman Coding

Name of the Innovative Practice: MIT Open Courseware Lecture Video

Date & Duration: 12-07-2019 & 20 minutes

ICT Tool Used:- Desktop, Projector

Description of Lecture Video:

A video lecture is a video which presents educational material for a topic which is to be learned. It might be a video of a teacher speaking to the camera, photographs and text about the topic or some mixture of these.

Goals(Learning Outcomes)

- The Students will be able to find the solution for the given problem statement.
- The Students will be able to generate shorter binary codes for encoding symbols/characters that appear more frequently in the input string

Use of appropriate methods:

Justification for choosing the topic using Lecture Video:

Studies have shown that the use of short video clips allows for more efficient processing and memory recall. The visual and auditory nature of videos appeals to a wide audience and allows each user to process information in a faster way to them. Also this topic was very important and very essential to know about how to generate the code words for different possible messages. So I chose this topic as a lecture video for better understanding and possible approaches to solve the problem.

Effective Presentation

Details of Implementation

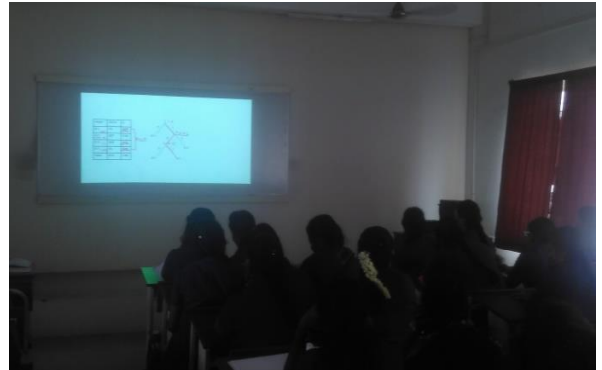
The Students watched the entire lecture video of how to generate the code words using Huffman coding and solve the problems.

Execution

Huffman Coding – MIT Lecture Video



MIT open Courseware Lecture Video on Huffman Coding



Students watching MIT Lecture Video

Significance of Results:

Assessment of Effectiveness/Success of the Activity:

The success of the activity was evaluated by asking the Huffman coding in Internal Assessment test I – Around 90% of students answered correctly.

Reflective Critique:

Benefits of conducting this activity

After Completion of the video, students gave feedback that they got a clear idea about how to encode the messages using Huffman coding, and they asked me to conduct more such activities like this.

Challenges

For better Clarity, I repeat some portion of the videos again to watch. Also the video quality was not much good.

Changes required for future (if required):

NIL

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	2				2		2	1	2

CO1: The Students will be able to design an application using various source coding techniques.

References:

1. S. Haykin, "Digital Communications", John Wiley, 2005
2. J.G Proakis, "Digital Communication", 4th Edition, Tata Mc Graw Hill Company, 2001
3. <https://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-004-computation-structures-spring-2017/c1/c1s2/c1s2v12/worked-example-huffman-encoding/>

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UNIT II WAVEFORM CODING

Degree, Semester& Branch: V Semester B.E. ECE B

Course Code & Title: EC8501 Digital Communication

Name of the Faculty member: Mr.R.Deiva Nayagam

Name of the Topic: Delta Modulation

Name of the Innovative Practice: Simulation using MATLAB

Date & Duration: 26-07-2019 & 20 minutes

ICT Tool Used:- Desktop, MATLAB tool, Communication Systems Laboratory

Description of Simulation:

Simulation is the imitation of the operation of a real-world process or system over time. The act of simulating something first requires that a model be developed; this model represents the key characteristics or behaviors of the selected physical or abstract system or process. The model represents the system itself, whereas the simulation represents the operation of the system over time.

Goals(Learning Outcomes)

- The Students will be able to know how the signal was modulated and transmitted.
- The Students will be able to change the parameter values and see what happens

Use of appropriate methods:

Justification for choosing Simulation:

Simulation allows students to change parameter values and see what happens. I chose this particular topic as a MATLAB simulation, because students studied the same concepts in theory. So they got better idea while implementing the same using MATLAB. Due to this activity, the students were able to simulate the original information into modulated data using delta modulation

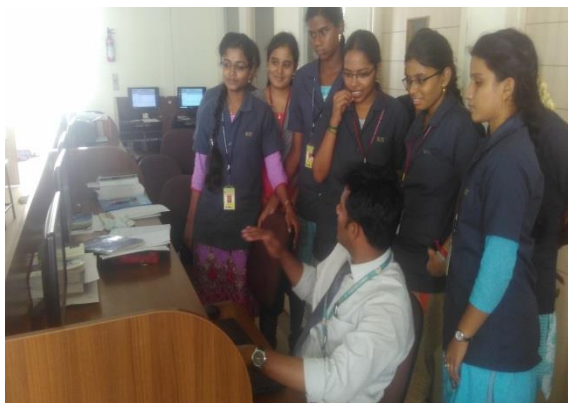
Effective Presentation

Details of Implementation

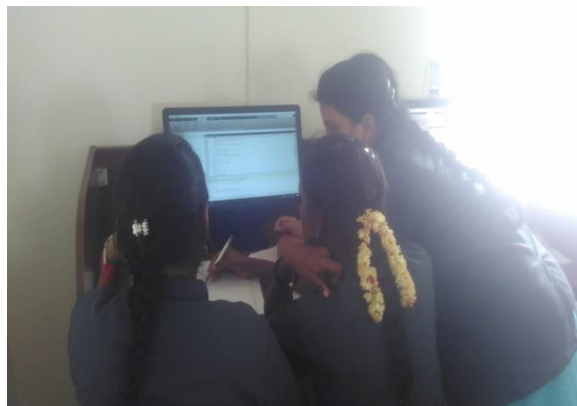
Students simulated the Delta modulation and demodulation using MATLAB tool.

Execution

Delta Modulation Simulation using MATLAB



Simulation of Delta Modulation was explained



Students Simulated Delta Modulation using MATLAB tool

Significance of Results:

Assessment of Effectiveness/Success of the Activity:

The assessment of effectiveness of the activity was felt when asked Viva questions after Simulation.

Reflective Critique:

Benefits of conducting this activity

After Completion of Simulation, the laboratory experiences will be more likely to achieve the goals with clear learning outcomes in mind. As they studied the concepts as a physical appearance, they were well versed in the particular topic. The laboratory Simulations are thoughtfully sequenced into the flow of classroom instruction.

Challenges

Some students felt little bit difficult to understand the MATLAB codes. But after my explanation about MATLAB commands, they got clear idea.

Changes required for future (if required):

NIL

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO2	3	1	2	-	-	-	-	-	3	2	-	-

CO2: The students will be able to interpret the various waveform coding schemes.

References:

1. <https://in.mathworks.com/matlabcentral/fileexchange/30308-delta-modulation>
2. J.G Proakis, "Digital Communication", 4th Edition, Tata Mc Graw Hill Company, 2001
3. S. Haykin, "Digital Communications", John Wiley, 2005

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UNIT III BASEBAND TRANSMISSION AND RECEPTION

Degree, Semester& Branch: V Semester B.E. ECE B

Course Code & Title: EC8501 Digital Communication

Name of the Faculty member: Mr.R.Deiva Nayagam

Name of the Topic: Eye Pattern/Eye Diagram

Name of the Innovative Practice: NPTEL Lecture Video

Date & Duration: 22 -08-2019 & 20 minutes

ICT Tool Used:- Smart Class Room

Description of NPTEL lecture Video:

NPTEL is an acronym for National Programme on Technology Enhanced Learning which is an initiative by seven Indian Institutes of Technology (IIT Bombay, Delhi, Guwahati, Kanpur, Kharagpur, Madras and Roorkee) and Indian Institute of Science (IISc) for creating course contents in engineering and science. NPTEL provides E-learning through online Web and Video courses in Engineering, Science and humanities streams.

Goals(Learning Outcomes)

- The Students will be able to get a good idea of signal quality in the digital domain.
- The Students will be able to recognize the effects of distortion.

Use of appropriate methods:

Justification for choosing this topic NPTEL lecture video:

Studies have shown that the use of short video clips allows for more efficient processing and memory recall. The visual and auditory nature of videos appeals to a wide audience and allows each user to process information in a faster way to them. As this topic was very important to analyse about how the digital signals getting degraded. Professors from Various IIT's & IISc explained the topic from their vast experience of knowledge was outstanding.

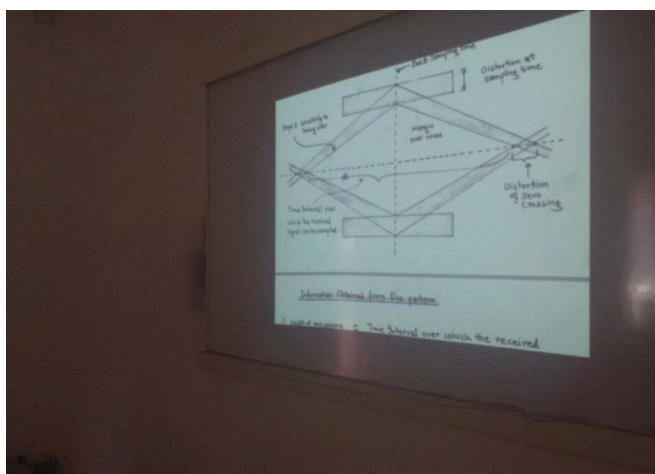
Effective Presentation

Details of Implementation

Students watched the NPTEL lecture video on EYE PATTERN.

Execution

Students watched NPTEL lecture video on EYE PATTERN



Significance of Results:

Assessment of Effectiveness/Success of the Activity:

The assessment of effectiveness of the activity was tested when asked in Internal Assessment Test. Around 80% of the students written well and good.

Reflective Critique:

Benefits of conducting this activity

The NPTEL lecture video will be more likely to understand than the other e-learning materials. The lecture video incorporates to understand the information in a faster way. It provokes thoughtfully sequenced into the flow of next topic. The way of explaining with short video lectures allows them to recall the topic easily and score good marks.

Challenges

I have planned this topic for 20 minutes. But for better clarification, it took for about 30 minutes

Changes required for future (if required):

NIL

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO3	3	3	-	-	-	-	-	-	3	2	-	2

CO3: The students will be able to analyze the various baseband transmission schemes.

References:

1. <http://www.digimat.in/nptel/courses/video/117104127/L52.html>
2. https://nptel.ac.in/content/storage2/nptel_data3/html/mhrd/ict/text/117104127/lec52.pdf
3. S. Haykin, “Digital Communications”, John Wiley, 2005

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UNIT IV DIGITAL MODULATION SCHEME

Degree, Semester& Branch: V Semester B.E. ECE B

Course Code & Title: EC8501 Digital Communication

Name of the Faculty member: Mr.R.Deiva Nayagam

Name of the Topic: QAM (Quadrature Amplitude Modulation)

Name of the Innovative Practice: Animated Video

Date & Duration: 19-09-2019 & 15 minutes

ICT Tool Used:- Smart Class Room

Description of Animated Video:

Animated video is an engaging instrument that can help you tell your story more comprehensively. 65% of people are visual learners, and 90% of the information transmitted to the brain is visual. Animation evokes emotions that resonate with the viewers. It sparks conversations and makes your audience want to follow the story.

Goals(Learning Outcomes)

- The Students will be able to get a good idea of how the digital cable channels are encoded and transmitted via cable television providers.

Use of appropriate methods:

Justification for choosing this topic as animated video:

QAM is a digital television standard using quadrature amplitude modulation. It is the format by which digital cable channels are encoded and transmitted via cable television providers. QAM is used in a variety of communications systems such as Dial-up modems and WiFi. So it must be essential to understand the concept effectively. Studies have shown that the use of short video clips allows for more efficient processing and memory recall. The visual and auditory nature of videos appeals to a wide audience and allows each user to process information in a faster way to them. So I chose this topic as an animated video.

Effective Presentation

Details of Implementation

Students watched the animated video on QAM signal and they got a better clarity on this topic.

Execution

Students watched Animated video on “QAM- Amplitude & Phase Shift keying”



Significance of Results:

Assessment of Effectiveness/Success of the Activity:

The assessment of effectiveness of the activity was tested when asked in Internal Assessment Test and in laboratory classes during viva-voce.

Reflective Critique:

Benefits of conducting this activity

The Animated video will be more likely to understand than the other e-learning materials. The animated video incorporates to understand the information in a faster way. The way of explanation with animation allows them to understand the concepts effectively and scored good marks in Internal Assessment Test.

Challenges

NIL

Changes required for future (if required):

NIL

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO4	3	2	1	-	-	-	-	1	3	2	-	2

CO4: The Students will be able to develop applications using the various band pass signalling schemes

References:

1. <https://www.youtube.com/watch?v=6BIqEWEE5-I>
2. <https://www.youtube.com/watch?v=d7l5NbFfBiU>
3. S. Haykin, "Digital Communications", John Wiley, 2005

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UNIT V ERROR CONTROL CODING

Degree, Semester& Branch: V Semester B.E. ECE B

Course Code & Title: EC8501 Digital Communication

Name of the Faculty member: Mr.R.Deiva Nayagam

Name of the Topic: Convolution Coding

Name of the Innovative Practice: MIT Lecture Video

Date & Duration: 03-10-2019 & 20 minutes

ICT Tool Used:- Smart Class Room

Description of Lecture Video:

A video lecture is a video which presents educational material for a topic which is to be learned. It might be a video of a teacher speaking to the camera, photographs and text about the topic or some mixture of these.

Goals(Learning Outcomes)

- The Students will be able to encode and decode a continuous stream of bits
- The Students will be able to design the system to detect, prevent and correct the errors during transmission.

Use of appropriate methods:

Justification for choosing the topic using Lecture Video:

Studies have shown that the use of short video clips allows for more efficient processing and memory recall. The visual and auditory nature of videos appeals to a wide audience and allows each user to process information in a faster way to them. So I chose this topic as a lecture video, because it is important to know about how to encode and decode the continuous stream of bits. Because of the nature of convolution coding technique the binary bit stream source is convolved by applying some binary operations on them.

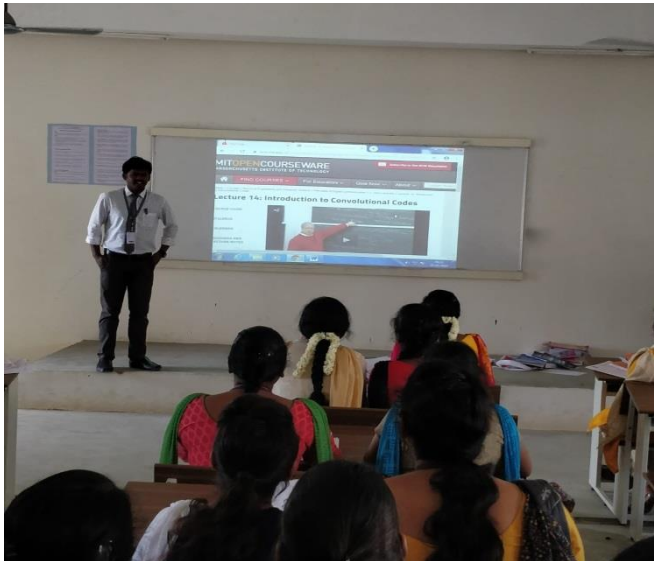
Effective Presentation

Details of Implementation

The Students watched the entire lecture video of how to encode and decode the signal using Convolution codes.

Execution

Students watched MIT Lecture Video on “Convolution codes”



Significance of Results:

Assessment of Effectiveness/Success of the Activity:

The success of the activity was evaluated by asking this topic in Internal Assessment test III Examinations. Most of them were answered correctly.

Reflective Critique:

Benefits of conducting this activity

After Completion of the video, students gave feedback that they got a clear idea about how to encode and decode the message using Convolution coding, and they asked me to conduct more such activities like this.

Challenges

For better Clarity, I replayed the video again as by the student requisition.

Changes required for future (if required):

NIL

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO5	3	3	2	2	-	1	-	-	-	2	-	-

CO5: The Students will be able to apply the basic concepts of channel coding

References:

1. S. Haykin, "Digital Communications", John Wiley, 2005
2. J.G Proakis, "Digital Communication", 4th Edition, Tata Mc Graw Hill Company, 2001
3. <https://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-451-principles-of-digital-communication-ii-spring-2005/video-lectures/lecture-13-introduction-to-convolutional-codes/>
4. <https://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-451-principles-of-digital-communication-ii-spring-2005/video-lectures/lecture-14-introduction-to-convolutional-codes/>